FURTHER MECHANICS / MOMENTUM & COLLISIONS

# Collisions in 2 Dimensions

Questions by topic 12 questions · 126 marks

EXAM BOARDS Edexcel FM1 9FM0/3C · AQA 7367/3M · OCR A Y543 · OCR MEI Mech Major Y421 CIE 9231 Paper 3

**Instructions**

Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question gives a value. Where a question leaves $g$ unspecified and a final answer depends on it, the solutions also give that answer for $g = 10\,\mathrm{m\,s^{-2}}$.

CIE (9231 Paper 3): papers take $g = 10\,\mathrm{m\,s^{-2}}$; use the $g = 10$ final answers where the solutions give them.

## Questions


| | | |
|---|---|---|
| Question 1 | *6 marks* | 2 |
| Question 2 | *11 marks* | 3 |
| Question 3 | *5 marks* | 4 |
| Question 4 | *9 marks* | 5 |
| Question 5 | *10 marks* | 6 |
| Question 6 | *9 marks* | 7 |
| Question 7 | *11 marks* | 8 |
| Question 8 | *11 marks* | 9 |
| Question 9 | *11 marks* | 10 |
| Question 10 | *14 marks* | 11 |
| Question 11 | *14 marks* | 12 |
| Question 12 | *15 marks* | 13 |

**1.** A hockey puck moving on a smooth horizontal rink collides with a fixed straight side-board.
Let $\mathbf{i}$ be a unit vector parallel to the side-board and let $\mathbf{j}$ be a unit vector perpendicular to the side-board, directed away from it.

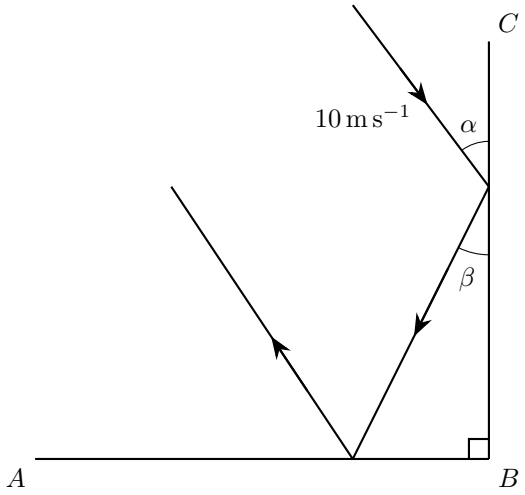
Immediately before the collision, the puck has velocity $(9\mathbf{i} - 12\mathbf{j})\,\mathrm{m\,s^{-1}}$.

Immediately after the collision, the puck moves with speed $10\,\mathrm{m\,s^{-1}}$ in the direction of the vector $3\mathbf{i} + 4\mathbf{j}$.

Model the puck as a particle.

**(a)** Find the coefficient of restitution between the puck and the side-board. **[3]**

**(b)** Determine whether or not the side-board is smooth.
Fully justify your answer. **[3]**




**2.** The diagram above represents the plan view of one corner of a smooth air-hockey table, where $AB$ and $BC$ are fixed cushions with $AB$ perpendicular to $BC$.

A small puck is struck along the table towards $BC$ with speed $10\,\mathrm{m\,s^{-1}}$ on a path that makes an angle $\alpha$ with $BC$, where $\tan\alpha = \frac{3}{4}$. The puck hits $BC$ and then hits $AB$.

Immediately after hitting $BC$, the puck is moving at an angle $\beta$ to $BC$, where $\tan\beta = \frac{1}{2}$.

The coefficient of restitution between the puck and $BC$ is $e$.

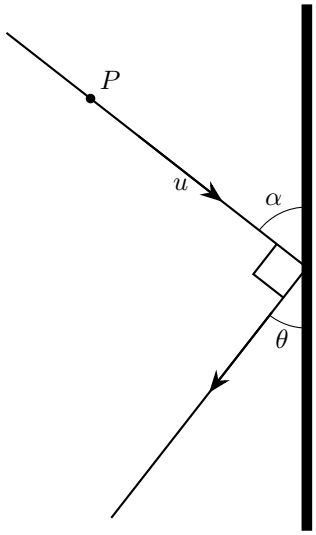
The coefficient of restitution between the puck and $AB$ is $\frac{3}{4}$.

By modelling the puck as a particle and the table and cushions as being smooth,

**(a)** show that $e = \frac{2}{3}$ **[5]**

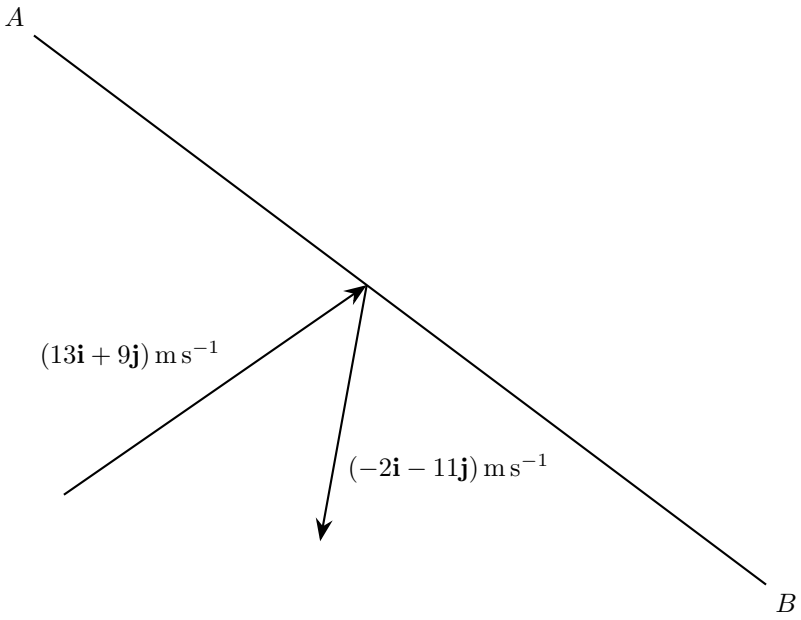
**(b)** find the speed of the puck immediately after it hits $AB$. **[4]**

**(c)** Suggest two ways in which the model could be refined to make it more realistic. **[2]**




**3.** A particle $P$ of mass $m$ is moving with speed $u$ on a fixed smooth horizontal surface. It collides with a fixed smooth vertical barrier. Before impact its path makes an angle $\alpha$ with the barrier, and after impact its path makes an angle $\theta$ with the barrier. The incident path and the outgoing path are perpendicular. The coefficient of restitution between $P$ and the barrier is $e$. The particle loses 36% of its kinetic energy in the collision.

Find the value of $e$ and the exact value of $\tan\alpha$. **[5]**




**4.** The diagram above shows the plan view of part of a smooth horizontal table. The line segment $AB$ represents a fixed smooth vertical wall.

A small ball of mass 0.4 kg moves across the table and rebounds after striking the wall.

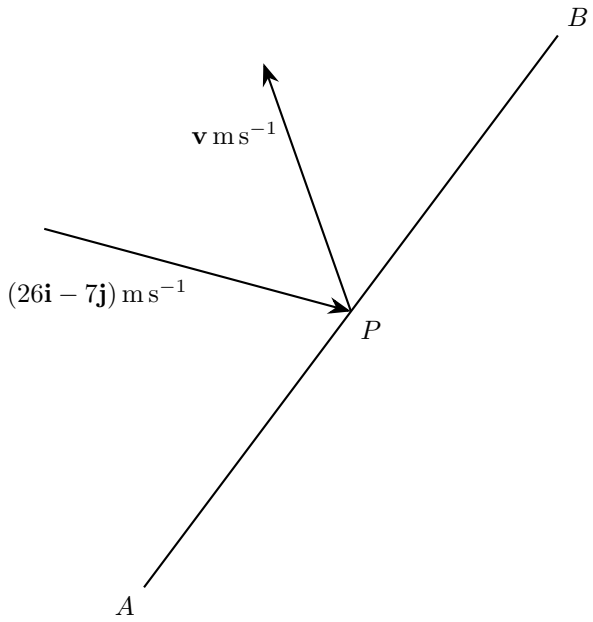
Immediately before impact its velocity is $(13\mathbf{i} + 9\mathbf{j})\,\mathrm{m\,s^{-1}}$.

Immediately after impact its velocity is $(-2\mathbf{i} - 11\mathbf{j})\,\mathrm{m\,s^{-1}}$.

The coefficient of restitution between the ball and the wall is $e$.

**(a)** Show that $AB$ is parallel to $(4\mathbf{i} - 3\mathbf{j})$. **[4]**

**(b)** Find the value of $e$. **[5]**




**5.** The diagram above represents the plan view of part of a smooth horizontal billiards table, where $AB$ is a fixed smooth cushion.

The direction of $\overrightarrow{AB}$ is parallel to the vector $(3\mathbf{i}+4\mathbf{j})$.

A small disc of mass 0.2 kg is moving on the table when it strikes the cushion $AB$.

Immediately before impact with $AB$, the velocity of the disc is $(26\mathbf{i}-7\mathbf{j})\,\mathrm{m\,s^{-1}}$.

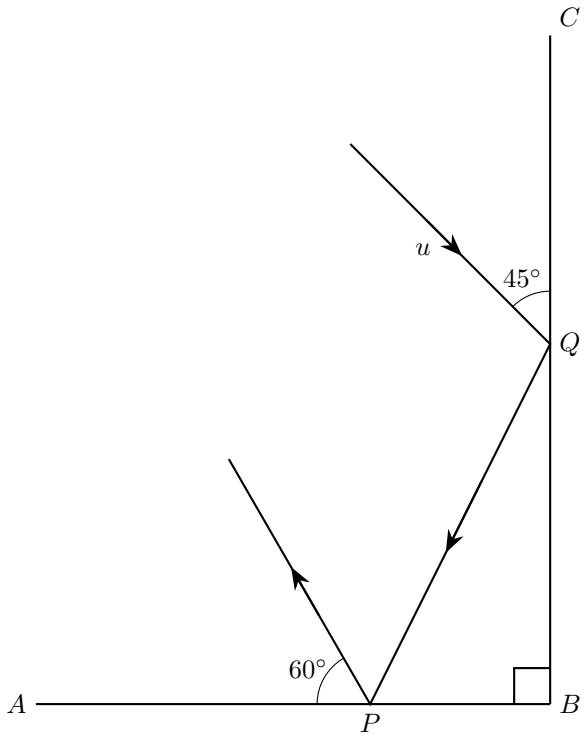
Immediately after impact with $AB$, the velocity of the disc is $\mathbf{v}\,\mathrm{m\,s^{-1}}$.

The coefficient of restitution between the disc and the cushion is $\frac{3}{5}$.

By modelling the disc as a particle,

**(a)** show that $\mathbf{v}=-6\mathbf{i}+17\mathbf{j}$ **[6]**

**(b)** Find the magnitude of the impulse exerted by the cushion on the disc in the impact. **[4]**




**6.** The diagram above represents the plan view of part of a horizontal floor, where $AB$ and $BC$ are perpendicular vertical walls.
The floor and the walls are modelled as smooth.

A ball is projected along the floor towards $BC$ with speed $u\,\mathrm{m\,s^{-1}}$ on a path at an angle of 45° to $BC$. The ball hits $BC$ at $Q$ and then hits $AB$ at $P$. After striking $AB$, the ball moves away on a path at an angle of 60° to $AB$.

The ball is modelled as a particle.

The coefficient of restitution between the ball and wall $BC$ is $\frac{1}{2}$.

**(a)** Find the coefficient of restitution between the ball and wall $AB$ and hence show that, using this model, the final kinetic energy of the ball is 50% of the initial kinetic energy of the ball. **[8]**

**(b)** In practice the walls and the floor may not be smooth. Explain how this model is likely to affect the percentage of kinetic energy remaining. **[1]**

**7.** A particle $P$ is moving with velocity $(7\mathbf{i} - 4\mathbf{j})\,\mathrm{m\,s^{-1}}$ on a smooth horizontal plane. It collides with a smooth vertical wall which is parallel to the direction vector $2\mathbf{i}+\mathbf{j}$ and rebounds with velocity $(2\mathbf{i}+6\mathbf{j})\,\mathrm{m\,s^{-1}}$.

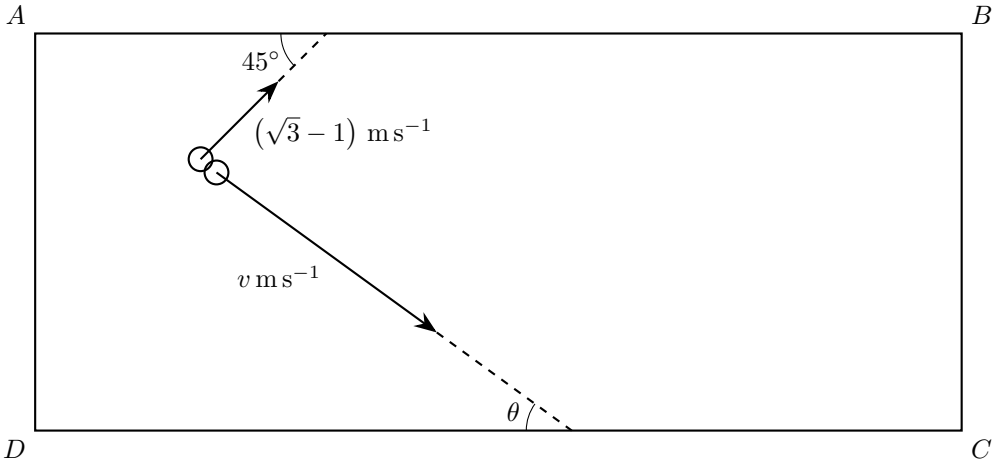
The coefficient of restitution between $P$ and this wall is $e$.

**(a)** Find the value of $e$. **[5]**

After this collision, $P$ goes on to hit a second smooth vertical wall which is parallel to the direction vector $\mathbf{i} - 2\mathbf{j}$. The coefficient of restitution between $P$ and this second wall is $\frac{1}{2}$.

The angle through which the direction of motion of $P$ is deflected by its collision with this second wall is $\alpha^\circ$.

**(b)** Find the value of $\alpha$, giving your answer to the nearest whole number. **[6]**




**8.** Two snooker balls, one white and one red, have equal mass.
The balls are on a horizontal table $ABCD$.

The white ball is struck so that it moves at a speed of $2\,\mathrm{m\,s^{-1}}$ in a direction making an angle of $15°$ with $AB$, towards $CD$.

The white ball hits a stationary red ball.

After the collision, the white ball moves at a speed of $(\sqrt{3}-1)\,\mathrm{m\,s^{-1}}$ and at an angle of $45°$ to $AB$.

After the collision, the red ball moves at a speed $v\,\mathrm{m\,s^{-1}}$ and at an angle $\theta$ to $CD$.

When the collision takes place, the point of impact is twice as far from $CD$ as it is from $AB$.

The diagram above shows the velocities of the balls after the collision.

After the collision, the white ball hits $AB$ and the red ball hits $CD$.

Model the balls as particles and neglect any resistance.

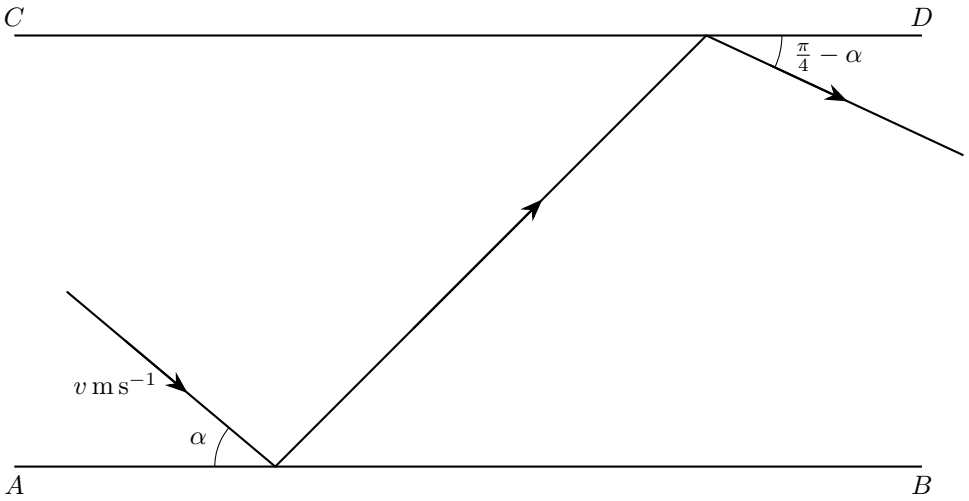
**(a)** Explain why the two balls hit the sides of the table at the same time. **[2]**

**(b)** Show that $\theta = 36.2°$ correct to one decimal place. **[4]**

**(c)** Find $v$. **[2]**

**(d)** Determine which ball travels the greater distance after the collision and before hitting the side of the table.
Fully justify your answer. **[2]**

**(e)** State one possible refinement to the model that you have used. **[1]**




**9.** The diagram above represents the plan view of part of a horizontal floor, where $AB$ and $CD$ represent fixed vertical walls, with $AB$ parallel to $CD$.

A small ball is projected along the floor towards wall $AB$. Immediately before hitting wall $AB$, the ball is moving with speed $v\,\mathrm{m\,s^{-1}}$ at an angle $\alpha$ to $AB$, where $0 < \alpha < \frac{\pi}{4}$.

The ball hits wall $AB$ and then hits wall $CD$.

Immediately after the impact with wall $CD$, the ball is moving at an angle $\frac{\pi}{4} - \alpha$ to $CD$.

The coefficient of restitution between the ball and wall $AB$ is $\frac{3}{5}$.

The coefficient of restitution between the ball and wall $CD$ is $\frac{1}{2}$.

The floor and the walls are modelled as being smooth. The ball is modelled as a particle.

**(a)** Show that

$$\tan\alpha = \frac{2}{3}$$ **[7]**

**(b)** Find the percentage of the ball's initial kinetic energy that is lost in the two impacts. **[4]**

**10.** A particle $P$ of mass $m$ is moving downwards and to the right in a direction making an angle $45°$ with the horizontal when it strikes a fixed smooth inclined plane. The plane is inclined to the horizontal at an angle $\alpha$, where $0 < \alpha < 45°$.

At the instant immediately before the impact, the speed of $P$ is $u$.

At the instant immediately after the impact, $P$ is moving upwards and to the right in a direction making an angle $45°$ with the horizontal, with speed $v$.

**(a)** Show that the magnitude of the impulse exerted on the plane by $P$ is $mu\sec(45° - \alpha)$. **[5]**

The coefficient of restitution between $P$ and the plane is $e$, where $0 < e < 1$.

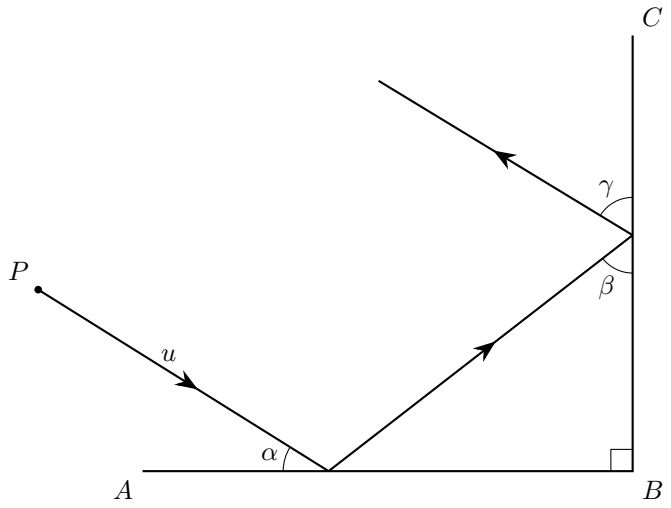
**(b)** Show that **[3]**

$$v^2 = u^2\left(\cos^2(45° + \alpha) + e^2\sin^2(45° + \alpha)\right)$$

**(c)** Show that the kinetic energy lost by $P$ in the impact is **[2]**

$$\frac{1}{2}mu^2(1 - e^2)\sin^2(45° + \alpha)$$

**(d)** Hence find, in terms of $m$, $u$ and $e$ only, the kinetic energy lost by $P$ in the impact. **[4]**




**11.** The smooth walls $AB$ and $BC$ are at right angles to each other. A particle $P$ moves on a smooth horizontal floor with speed $u$ and strikes the wall $AB$ at an angle $\alpha$ to $AB$. It rebounds, then strikes the wall $BC$ at an angle $\beta$ to $BC$. After the second impact it rebounds at an angle $\gamma$ to $BC$, as shown in the diagram. The coefficient of restitution between $P$ and each wall is $e$.

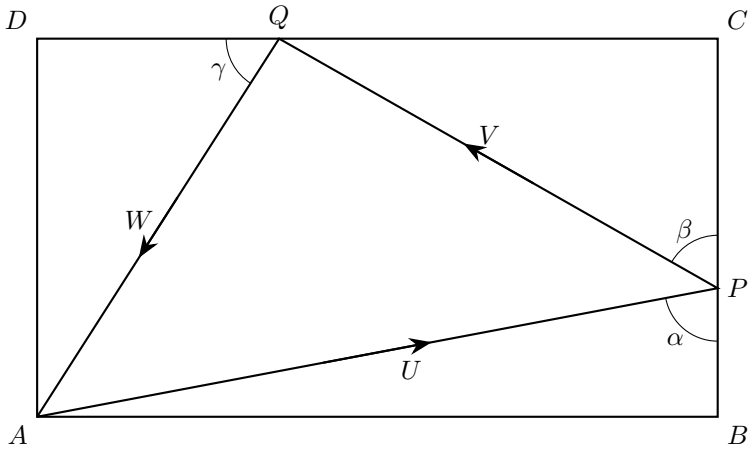
**(a)** Show that

$$\tan\alpha \tan\beta = \frac{1}{e}$$ **[4]**

**(b)** Express $\gamma$ in terms of $\alpha$ and explain what this implies about the final direction of motion of $P$. **[4]**

During the first impact the particle loses four times as much kinetic energy as it loses during the second impact. After the second impact the kinetic energy of $P$ is $\frac{9}{25}$ of its initial kinetic energy.

**(c)** Find the value of $e$ and the value of $\tan\alpha$. **[6]**




**12.** A small smooth billiard ball is projected from the corner $A$ of a horizontal rectangular table $ABCD$.

The ball first hits the side $BC$ at the point $P$, then hits the side $CD$ at the point $Q$, and then returns to $A$.

Angle $APB = \alpha$, angle $CPQ = \beta$ and angle $DQA = \gamma$.

The ball moves along $AP$ with speed $U$, along $PQ$ with speed $V$ and along $QA$ with speed $W$, as shown in the diagram.

The coefficient of restitution between the ball and side $BC$ is $e_1$.

The coefficient of restitution between the ball and side $CD$ is $e_2$.

The ball is modelled as a particle.

**(a)** Show that $\tan\beta = e_1 \tan\alpha$. **[4]**

**(b)** Hence show that $e_1 \tan\alpha \tan\gamma = e_2$. **[4]**

**(c)** By considering $\angle PAQ$ or otherwise, show that it is possible for the ball to return to $A$ only if $e_2 > e_1$. **[6]**

If instead $e_1 = e_2$, the ball would not return to $A$.

**(d)** Given that $e_1 = e_2$, use the result from part (b) to describe the path of the ball after it hits $CD$ at $Q$, explaining your answer. **[1]**